# Introduction to Particle Physics

To begin with, we must decide what physical quantities we would like to calculate. In the case of decays, the item of greatest interest is the *lifetime* of the particle in question. What precisely do we mean by the lifetime of, say, the muon? We have in mind, of course, a muon at *rest*; a moving muon lasts longer (from our perspective) because of time dilation. But even stationary muons don't all last the same amount of time, for there is an intrinsically random element in the decay process. We cannot hope to calculate the lifetime of any *particular* muon; rather, what we are after is the *average* (or 'mean') lifetime, $\tau$, of the muons in any large sample.

Now, elementary particles have no memories, so the probability of a given muon decaying in the next microsecond is independent of how long ago that muon was created. (It's quite different in biological systems: an 80-year-old man is much more likely to die in the next year than is a 20-year-old, and his body shows the signs of eight decades of wear and tear. But all muons are identical, regardless of when they were produced; from an actuarial point of view they're all on an equal footing.) The critical parameter, then, is the *decay rate*, $\Gamma$, the *probability per unit time* that any given muon will disintegrate. If we had a large collection of muons, say, $N(t)$, at time $t$, then $N\Gamma dt$ of them would decay in the next instant $dt$. This would, of course, *decrease* the number remaining:

$$dN = -\Gamma N \, dt$$

It follows that

$$N(t) = N(0)e^{-\Gamma t}$$

Evidently, the number of particles left decreases exponentially with time.
the mean lifetime is simply the reciprocal of the decay rate:

$$\tau = \frac{1}{\Gamma}$$

Actually, most particles can decay by several different routes. The $\pi^+$, for instance, usually decays to $\mu^+ + \nu_\mu$, but sometimes one goes to $e^+ + \nu_e$; occasionally, a $\pi^+$ decays to $\mu^+ + \nu_\mu + \gamma$, and they have even been known to go to $e^+ + \nu_e + \pi^0$. In such circumstances, the *total* decay rate is the sum of the individual decay rates:

$$\Gamma_{\text{tot}} = \sum_{i=1}^{n} \Gamma_i$$

and the lifetime of the particle is the reciprocal of $\Gamma_{\text{tot}}$:

$$\tau = \frac{1}{\Gamma_{\text{tot}}}$$

In addition to $\tau$, we want to calculate the various *branching ratios*, that is, the fraction of all particles of the given type that decay by each mode. Branching ratios are determined by the decay rates:

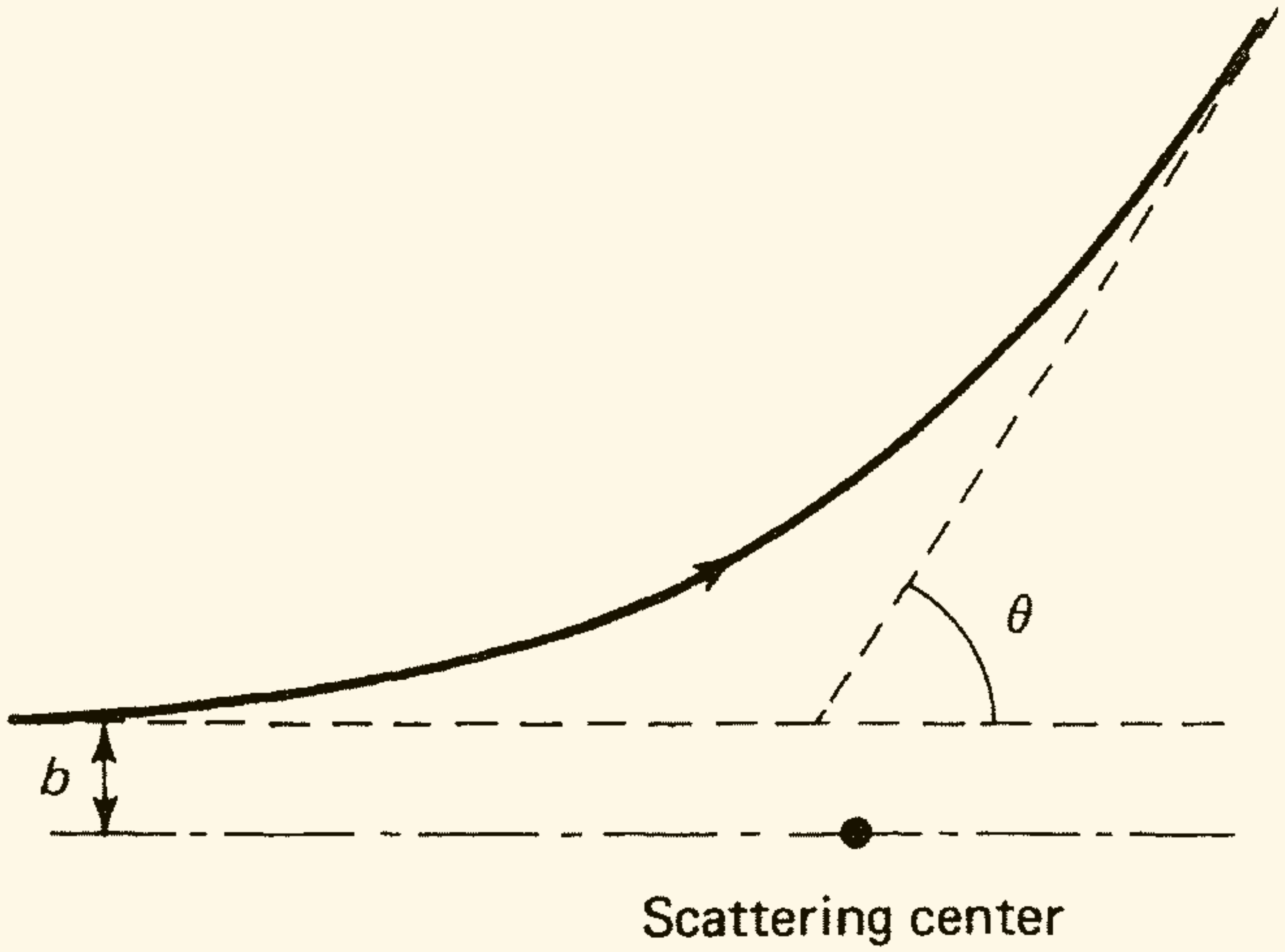
$$\text{Branching ratio for ith decay mode} = \Gamma_i / \Gamma_{\text{tot}}$$

For decays, then, the essential problem is to calculate the decay rate $\Gamma_i$ for each mode; from there it is an easy matter to obtain the lifetime and branching ratios.

How about scattering? What quantity should the experimentalist measure and the theorist calculate? If we were talking about an *archer* aiming at a 'bull's-eye', the parameter of interest would be the *size of the target* or, more precisely, the cross-sectional area it presents to a stream of incoming arrows. In a crude sense, the same goes for elementary particle scattering: if you fire a stream of electrons into a tank of hydrogen (which is essentially a collection of protons), the parameter of interest is the size of the proton – the cross-sectional area $\sigma$ it presents to the incident beam. The situation is more complicated than in archery, however, for several reasons. First of all the target is 'soft'; it's not a simple case of 'hit-or-miss', but rather 'the closer you come the greater the deflection'. Nevertheless, it is still possible to define an 'effective' cross section; I'll show you how in a moment. Secondly, the cross section depends on the nature of the 'arrow' as well as the structure of the 'target'. Electrons scatter off hydrogen more sharply than neutrinos and less so than pions, because different interactions are involved. It depends, too, on the *outgoing* particles; if the energy is high enough we can have not only *elastic* scattering ($e + p \to e + p$), but also a variety of *inelastic* processes, such as $e + p \to e + p + \gamma$, or $e + p + \pi^0$, or even, in principle, $\nu_e + \Lambda$. Each one of these has its own ('exclusive') scattering cross section, $\sigma_i$ (for process $i$). In some experiments, however, the final products are not examined, and we are interested only in the *total* ('inclusive') cross section:

$$\sigma_{\text{tot}} = \sum_{i=1}^{n} \sigma_i$$

Finally, each cross section typically depends on the *velocity* of the incident particle. At the most naive level we might expect the cross section to be proportional to the amount of time the incident particle spends in the vicinity of the target, which is to say that $\sigma$ should be inversely proportional to $v$. But this behavior is dramatically altered in the neighborhood of a 'resonance' – a special energy at which the particles involved 'like' to interact, forming a short-lived semibound state before breaking apart. Such 'bumps' in the graph of $\sigma$ versus $v$ (or, as it is more commonly plotted, $\sigma$ versus $E$) are in fact the principal means by which short-lived particles are discovered (see Figure ). So, unlike the archer's target, there's a lot of physics in an elementary particle cross section.

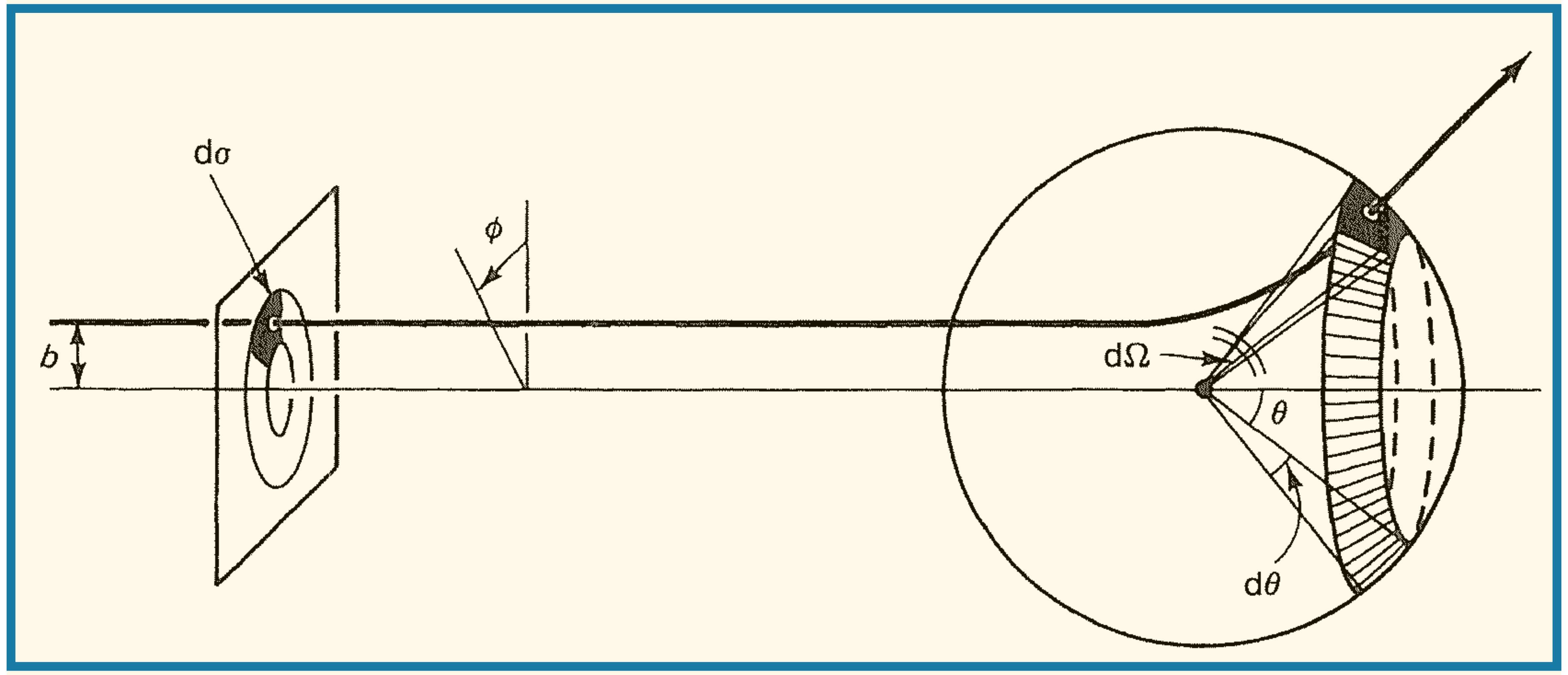


# The Feynman Calculus

## Scattering Cross Sections

If the particle comes in with an impact parameter between $b$ and $b + db$, it will emerge with a scattering angle between $\theta$ and $\theta + d\theta$. More generally, if it passes through an infinitesimal *area* $d\sigma$, it will scatter into a corresponding *solid angle* $d\Omega$ . Naturally, the larger we make $d\sigma$, the larger $d\Omega$ will be. The proportionality factor is called the *differential (scattering) cross section, D*:*

$$d\sigma = D(\theta)\, d\Omega$$




The name is poorly chosen; it's not a differential, or even a derivative, in the mathematical sense. The words would apply more naturally to $d\sigma$ than to $d\sigma/d\Omega$ . . . but I'm afraid we're stuck with it.

Now,

$$d\sigma = |b\, db\, d\phi|, \quad d\Omega = |\sin\theta\, d\theta\, d\phi|$$

(Areas and solid angles are intrinsically positive, hence the absolute value signs.) Accordingly,

$$D(\theta) = \frac{d\sigma}{d\Omega} = \left|\frac{b}{\sin\theta}\left(\frac{db}{d\theta}\right)\right|$$

*Example 6.1 Hard-sphere Scattering* Suppose the particle bounces elastically sphere of radius $R$. we have

$$b = R\sin\alpha, \qquad 2\alpha + \theta = \pi$$

Thus,

$$\sin\alpha = \sin(\pi/2 - \theta/2) = \cos(\theta/2)$$

and hence

$$b = R\cos(\theta/2) \quad \text{or} \quad \theta = 2\cos^{-1}(b/R)$$

This is the relation between $\theta$ and $b$ for classical hard-sphere scattering.

$$\frac{db}{d\theta} = -\frac{R}{2}\sin\left(\frac{\theta}{2}\right)$$

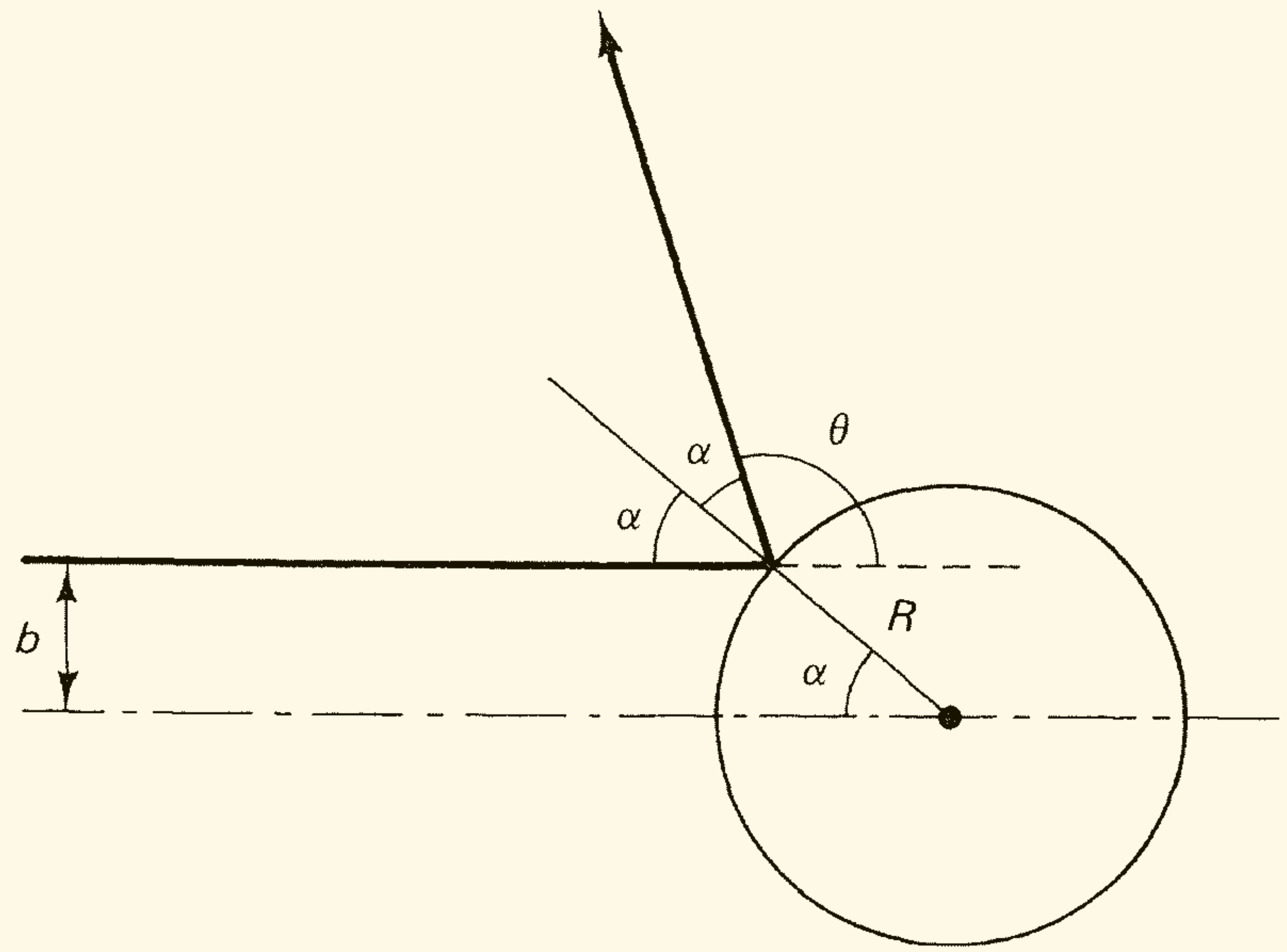
and hence

$$D(\theta) = \frac{Rb\sin(\theta/2)}{2\sin\theta} = \frac{R^2}{2}\frac{\cos(\theta/2)\sin(\theta/2)}{\sin\theta} = \frac{R^2}{4}$$

Finally, the *total* cross section is the integral of $d\sigma$ over all solid angles:

$$\sigma = \int d\sigma = \int D(\theta)\, d\Omega$$

$$\sigma = \int \frac{R^2}{4}\, d\Omega = \pi R^2$$



# The Feynman Calculus

## Scattering Cross Sections

Suppose we have a *beam* of incoming particles, with uniform *luminosity* $\mathcal{L}$ ($\mathcal{L}$ is the number of particles passing down the line per unit time, per unit area). Then $dN = \mathcal{L}\, d\sigma$ is the number of particles per unit time passing through area $d\sigma$, and hence also the number per unit time scattered into solid angle $d\Omega$:

$$dN = \mathcal{L}\, d\sigma = \mathcal{L} D(\theta)\, d\Omega$$

Suppose I set up a detector that subtends a solid angle $d\Omega$ with respect to the collision point I count the number of particles per unit time ($dN$) reaching my detector – what an experimentalist would call the *event rate*. the event rate is equal to the luminosity times the differential cross section times the solid angle. Whoever is operating the accelerator controls the luminosity; whoever set up the detector determined the solid angle. With these parameters established, the differential cross section can be measured by simply counting the number of particles entering the detector:

$$\frac{d\sigma}{d\Omega} = \frac{dN}{\mathcal{L}\, d\Omega}$$

If the detector completely surrounds the target, then $N = \sigma \mathcal{L}$; as accelerator physicists like to say, 'the event rate is the cross section times the luminosity'.

### Fermi's Golden Rule

The ritual for calculating reaction rates was dubbed the *Golden Rule* by Enrico Fermi. In essence, Fermi's Golden Rule says that a transition rate is given by the *product* of the phase space and the (absolute) square of the amplitude.

Suppose particle 1 (at rest) decays into several other particles 2, 3, 4, . . . , $n$:

$$1 \to 2 + 3 + 4 + \cdots + n$$

The decay rate is given by the formula

$$\Gamma = \frac{S}{2\hbar m_1} \int |\mathcal{M}|^2 (2\pi)^4 \delta^4 (p_1 - p_2 - p_3 \cdots - p_n) \times \prod_{j=2}^{n} 2\pi\, \delta\left(p_j^2 - m_j^2 c^2\right) \theta\left(p_j^0\right) \frac{d^4 p_j}{(2\pi)^4}$$

where $m_i$ is the mass of the $i$th particle and $p_i$ is its four-momentum. $S$ is a statistical factor that corrects for double-counting when there are identical particles in the final state: for each such group of $s$ particles, $S$ gets a factor of $(1/s!)$. For instance, if $a \to b + b + c + c + c$, then $S = (1/2!)(1/3!) = 1/12$. If there are *no* identical particles in the final state (the most common circumstance), then $S = 1$.

# The Feynman Calculus



## Fermi's Golden Rule

The *dynamics* of the process is contained in the *amplitude*, $\mathscr{M}(p_1, p_2, \ldots, p_n)$, which is a function of the various momenta; we'll calculate it (later) by evaluating the appropriate Feynman diagrams. The rest is *phase space*; it tells us to *integrate over all outgoing four-momenta*, subject to three kinematical constraints:

1. Each outgoing particle lies on its mass shell: $p_j^2 = m_j^2 c^2$ (which is to say, $E_j^2 - \mathbf{p}_j^2 c^2 = m_j^2 c^4$). This is enforced by the delta function $\delta\left(p_j^2 - m_j^2 c^2\right)$, which is zero unless its argument vanishes.*
2. Each outgoing energy is positive: $p_j^0 = E_j/c > 0$. Hence the $\theta$ function.†
3. Energy and momentum must be conserved: $p_1 = p_2 + p_3 \cdots + p_n$. This is ensured by the factor $\delta^4(p_1 - p_2 - p_3 \cdots - p_n)$.

The Golden Rule may *look* forbidding, but what it actually *says* could hardly be simpler: all outcomes consistent with the three natural kinematic constraints are a priori equally likely. To be sure, the *dynamics* (contained in $\mathscr{M}$) may favor some combinations of momenta over others, but with that modulation you just *add up all the possibilities*. How about all those factors of $2\pi$? These are easy to keep track of if you adhere scrupulously to the following rule:

Every $\delta$ gets $(2\pi)$; every d gets $1/(2\pi)$.

Four-dimensional 'volume' elements can be split into spatial and temporal parts:

$$d^4 p = dp^0 \, d^3\mathbf{p}$$

The $p^0$ integrals§ can be performed immediately, by exploiting the delta function

$$\delta\left(p^2 - m^2c^2\right) = \delta\left[(p^0)^2 - \mathbf{p}^2 - m^2c^2\right]$$

Now

$$\delta(x^2 - a^2) = \frac{1}{2a}\left[\delta(x-a) + \delta(x+a)\right] \quad (a > 0)$$

$$\theta\left(p^0\right)\delta\left[(p^0)^2 - \mathbf{p}^2 - m^2c^2\right] = \frac{1}{2\sqrt{\mathbf{p}^2 + m^2c^2}}\,\delta\left(p^0 - \sqrt{\mathbf{p}^2 + m^2c^2}\right)$$

(the theta function kills the spike at $p^0 = -\sqrt{\mathbf{p}^2 + m^2c^2}$, and it's 1 at $p^0 = \sqrt{\mathbf{p}^2 + m^2c^2}$). Thus

$$\Gamma = \frac{S}{2\hbar m_1}\int |\mathscr{M}|^2 (2\pi)^4 \delta^4\left(p_1 - p_2 - p_3 \cdots - p_n\right) \times \prod_{j=2}^{n} \frac{1}{2\sqrt{\mathbf{p}_j^2 + m_j^2c^2}} \frac{d^3\mathbf{p}_j}{(2\pi)^3}$$

with

$$p_j^0 \to \sqrt{\mathbf{p}_j^2 + m_j^2c^2}$$

wherever it appears (in $\mathscr{M}$ and in the remaining delta function). This is a more useful way to express the Golden Rule, though it obscures the physical content.

# The Feynman Calculus

## Scattering Cross Sections

### Two Particle Decays

In particular, if there are only two particles in the final state

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathscr{M}|^2 \frac{\delta^4(p_1 - p_2 - p_3)}{\sqrt{\mathbf{p}_2^2 + m_2^2 c^2}\sqrt{\mathbf{p}_3^2 + m_3^2 c^2}} \, d^3\mathbf{p}_2 \times d^3\mathbf{p}_3$$

The four-dimensional delta function is a product of temporal and spatial parts:

$$\delta^4(p_1 - p_2 - p_3) = \delta\left(p_1^0 - p_2^0 - p_3^0\right)\delta^3\left(\mathbf{p}_1 - \mathbf{p}_2 - \mathbf{p}_3\right)$$

But particle 1 is at rest, so $\mathbf{p}_1 = \mathbf{0}$ and $p_1^0 = m_1 c$. Meanwhile, $p_2^0$ and $p_3^0$ have been replaced

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathscr{M}|^2 \frac{\delta\left(m_1 c - \sqrt{\mathbf{p}_2^2 + m_2^2 c^2} - \sqrt{\mathbf{p}_3^2 + m_3^2 c^2}\right)}{\sqrt{\mathbf{p}_2^2 + m_2^2 c^2}\sqrt{\mathbf{p}_3^2 + m_3^2 c^2}}$$

$$\times\, \delta^3\left(\mathbf{p}_2 + \mathbf{p}_3\right)\, d^3\mathbf{p}_2\, d^3\mathbf{p}_3$$

The $\mathbf{p}_3$ integral is now trivial: in view of the final delta function it simply make: replacement

$$\mathbf{p}_3 \to -\mathbf{p}_2$$

leaving

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathscr{M}|^2 \frac{\delta\left(m_1 c - \sqrt{\mathbf{p}_2^2 + m_2^2 c^2} - \sqrt{\mathbf{p}_2^2 + m_3^2 c^2}\right)}{\sqrt{\mathbf{p}_2^2 + m_2^2 c^2}\sqrt{\mathbf{p}_2^2 + m_3^2 c^2}} \, d^3\mathbf{p}_2$$

For the remaining integral we adopt spherical coordinates, $\mathbf{p}_2 \to (r,\ \theta,\ \phi)$, $d^3\mathbf{p}_2 \to r^2 \sin\theta\, dr\, d\theta\, d\phi$ (this is *momentum* space, of course: $r = |\mathbf{p}_2|$).

$$\Gamma = \frac{S}{32\pi^2 \hbar m_1} \int |\mathscr{M}|^2 \frac{\delta\left(m_1 c - \sqrt{r^2 + m_2^2 c^2} - \sqrt{r^2 + m_3^2 c^2}\right)}{\sqrt{r^2 + m_2^2 c^2}\sqrt{r^2 + m_3^2 c^2}}$$

$$\times\, r^2 \sin\theta\, dr\, d\theta\, d\phi$$

Now, $\mathscr{M}$ was originally a function of the four-momenta $p_1$, $p_2$, and $p_3$, but $p_1 = (m_1 c, \mathbf{0})$ is a constant (as far as the integration is concerned), and the integrals already performed have made the replacements $p_2^0 \to \sqrt{\mathbf{p}_2^2 + m_2^2 c^2}$, $p_3^0 \to \sqrt{\mathbf{p}_3^2 + m_3^2 c^2}$, and $\mathbf{p}_3 \to -\mathbf{p}_2$, so by now $\mathscr{M}$ depends only on $\mathbf{p}_2$. As we shall see, however, amplitudes must be *scalars*, and the only scalar you can make out of a vector is the dot product with itself:* $\mathbf{p}_2 \cdot \mathbf{p}_2 = r^2$. At this stage, then, $\mathscr{M}$ is a function only of $r$ (not of $\theta$ or $\phi$). That being the case we can do the angular integrals

$$\int_0^{\pi} \sin\theta\, d\theta = 2, \qquad \int_0^{2\pi} d\phi = 2\pi$$

and there remains only the $r$ integral:

$$\Gamma = \frac{S}{8\pi \hbar m_1} \int_0^{\infty} |\mathscr{M}(r)|^2 \frac{\delta\left(m_1 c - \sqrt{r^2 + m_2^2 c^2} - \sqrt{r^2 + m_3^2 c^2}\right)}{\sqrt{r^2 + m_2^2 c^2}\sqrt{r^2 + m_3^2 c^2}} r^2\, dr$$

# The Feynman Calculus

## Scattering Cross Sections

### Two Particle Decays

To simplify the argument of the delta function, let

$$u \equiv \sqrt{r^2 + m_2^2c^2} + \sqrt{r^2 + m_3^2c^2}$$

so

$$\frac{\mathrm{d}u}{\mathrm{d}r} = \frac{ur}{\sqrt{r^2 + m_2^2c^2}\sqrt{r^2 + m_3^2c^2}}$$

Then

$$\Gamma = \frac{S}{8\pi\hbar m_1}\int_{(m_2+m_3)c}^{\infty} |\mathscr{M}(r)|^2 \delta\left(m_1 c - u\right)\frac{r}{u}\,\mathrm{d}u$$

The last integral sends* $u$ to $m_1c$, and hence $r$ to

$$r_0 = \frac{c}{2m_1}\sqrt{m_1^4 + m_2^4 + m_3^4 - 2m_1^2m_2^2 - 2m_1^2m_3^2 - 2m_2^2m_3^2}$$

Remember that $r$ was short for the variable $|\mathbf{p}_2|$; $r_0$ is the value of $|\mathbf{p}_2|$ that is consistent with conservation of energy,

$$\Gamma = \frac{S|\mathbf{p}|}{8\pi\hbar m_1^2 c}|\mathscr{M}|^2$$

where $|\mathbf{p}|$ is the magnitude of either outgoing momentum,

### Scattering

Suppose particles 1 and 2 collide, producing particles 3, 4, . . . , $n$:

$$1 + 2 \to 3 + 4 + \cdots + n$$

The scattering cross section is given by the formula

$$\sigma = \frac{S\hbar^2}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}}\int |\mathscr{M}|^2(2\pi)^4\delta^4(p_1 + p_2 - p_3 \cdots - p_n)$$
$$\times \prod_{j=3}^{n} 2\pi\,\delta\left(p_j^2 - m_j^2c^2\right)\theta\left(p_j^0\right)\frac{\mathrm{d}^4 p_j}{(2\pi)^4}$$

where $p_i$ is the four-momentum of particle $i$ (mass $m_i$) and the statistical factor ($S$) is the same as before . The phase space is essentially the same as before: integrate over all outgoing momenta, subject to the three kinematical constraints (every outgoing particle is on its mass shell, every outgoing energy is positive, and energy and momentum are conserved), which are enforced by the delta and theta functions. Once again, we can simplify matters by performing the $p_j^0$ integrals:

$$\sigma = \frac{S\hbar^2}{4\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2}}\int |\mathscr{M}|^2(2\pi)^4\delta^4(p_1 + p_2 - p_3 \cdots - p_n)$$
$$\times \prod_{j=3}^{n} \frac{1}{2\sqrt{\mathbf{p}_j^2 + m_j^2c^2}}\frac{\mathrm{d}^3\mathbf{p}_j}{(2\pi)^3}$$

# The Feynman Calculus



## Two-body Scattering in CM Frame

Consider the process

$$1 + 2 \rightarrow 3 + 4$$


in the CM frame, $\mathbf{p}_2 = -\mathbf{p}_1$

$$\sqrt{(p_1 \cdot p_2)^2 - (m_1 m_2 c^2)^2} = (E_1 + E_2)|\mathbf{p}_1|/c$$

$$\sigma = \frac{S\hbar^2 c}{64\pi^2 (E_1 + E_2)|\mathbf{p}_1|} \int |\mathscr{M}|^2 \frac{\delta^4(p_1 + p_2 - p_3 - p_4)}{\sqrt{\mathbf{p}_3^2 + m_3^2 c^2}\sqrt{\mathbf{p}_4^2 + m_4^2 c^2}} \, d^3\mathbf{p}_3 \, d^3\mathbf{p}_4$$

As before, we begin by rewriting the delta function:

$$\delta^4(p_1 + p_2 - p_3 - p_4) = \delta\left(\frac{E_1 + E_2}{c} - p_3^0 - p_4^0\right) \delta^3(\mathbf{p}_3 + \mathbf{p}_4)$$

and carry out the $\mathbf{p}_4$ integral (which sends $\mathbf{p}_4 \rightarrow -\mathbf{p}_3$):

$$\sigma = \left(\frac{\hbar}{8\pi}\right)^2 \frac{Sc}{(E_1 + E_2)|\mathbf{p}_1|} \int |\mathscr{M}|^2 \times \frac{\delta\left[(E_1 + E_2)/c - \sqrt{\mathbf{p}_3^2 + m_3^2 c^2} - \sqrt{\mathbf{p}_3^2 + m_4^2 c^2}\right]}{\sqrt{\mathbf{p}_3^2 + m_3^2 c^2}\sqrt{\mathbf{p}_3^2 + m_4^2 c^2}} \, d^3\mathbf{p}_3$$

This time, however, $|\mathscr{M}|^2$ depends on the *direction* of $\mathbf{p}_3$ as well as its magnitude, so we cannot carry out the angular integration. But that's all right – we didn't really want $\sigma$ in the first place; what we're after is $d\sigma/d\Omega$. Adopting spherical coordinates, as before,

$$d^3\mathbf{p}_3 = r^2 \, dr \, d\Omega$$

(where $r$ is shorthand for $|\mathbf{p}_3|$ and $d\Omega = \sin\theta \, d\theta \, d\phi$), we obtain

$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar}{8\pi}\right)^2 \frac{Sc}{(E_1 + E_2)|\mathbf{p}_1|} \int_0^\infty |\mathscr{M}|^2 \times \frac{\delta\left[(E_1 + E_2)/c - \sqrt{r^2 + m_3^2 c^2} - \sqrt{r^2 + m_4^2 c^2}\right]}{\sqrt{r^2 + m_3^2 c^2}\sqrt{r^2 + m_4^2 c^2}} \, r^2 \, dr$$

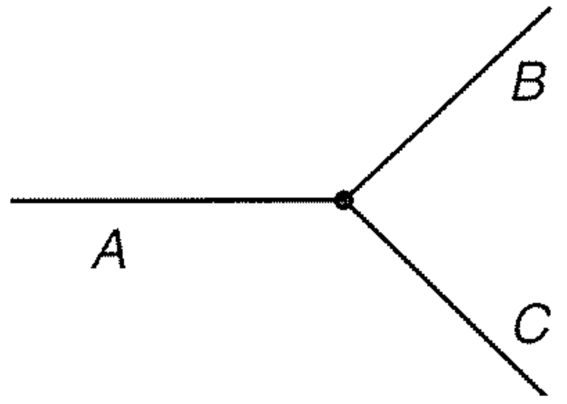
$$\frac{d\sigma}{d\Omega} = \left(\frac{\hbar c}{8\pi}\right)^2 \frac{S|\mathscr{M}|^2}{(E_1 + E_2)^2} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}$$

where $|\mathbf{p}_f|$ is the magnitude of either outgoing momentum and $|\mathbf{p}_i|$ is the magnitude of either incoming momentum.

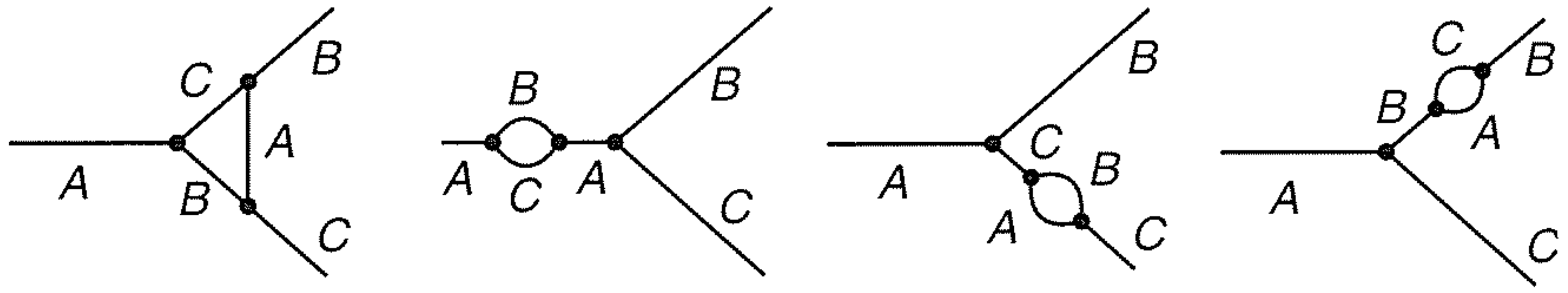
# The Feynman Calculus

## Feynman Rules

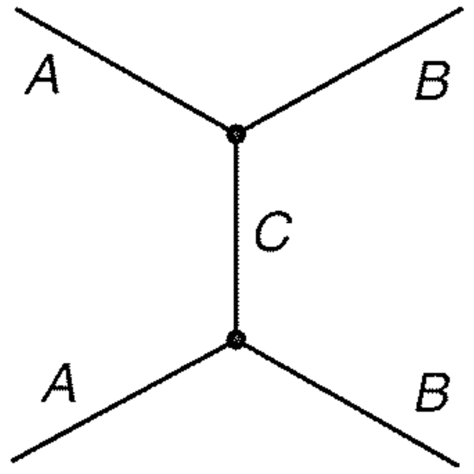
Imagine a world in which there are just three kinds of particles – call them $A$, $B$, and $C$ – with masses $m_A$, $m_B$, and $m_C$. They all have spin 0 and each is its own antiparticle (so we don't need arrows on the lines). There is one primitive vertex, by which the three particles interact:



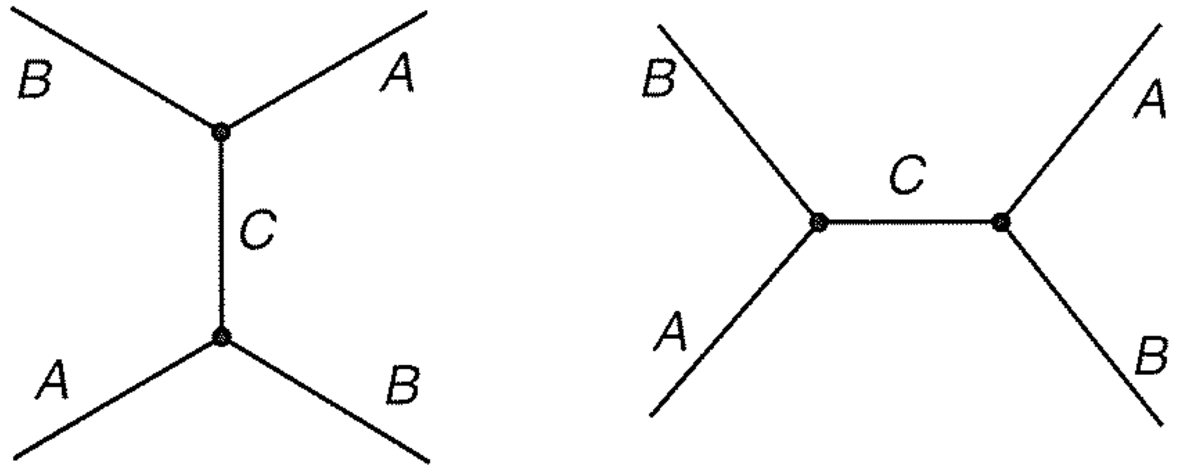

I shall assume that $A$ is the heaviest of the three and in fact weighs more than $B$ and $C$ combined, so that it can decay into $B + C$. The lowest-order diagram describing this disintegration is the primitive vertex itself; to this there are (small) third-order corrections:




and even smaller ones of higher order. Our first project will be to calculate the lifetime of the $A$, to lowest order. After that, we'll look at various scattering processes, such as $A + A \to B + B$:



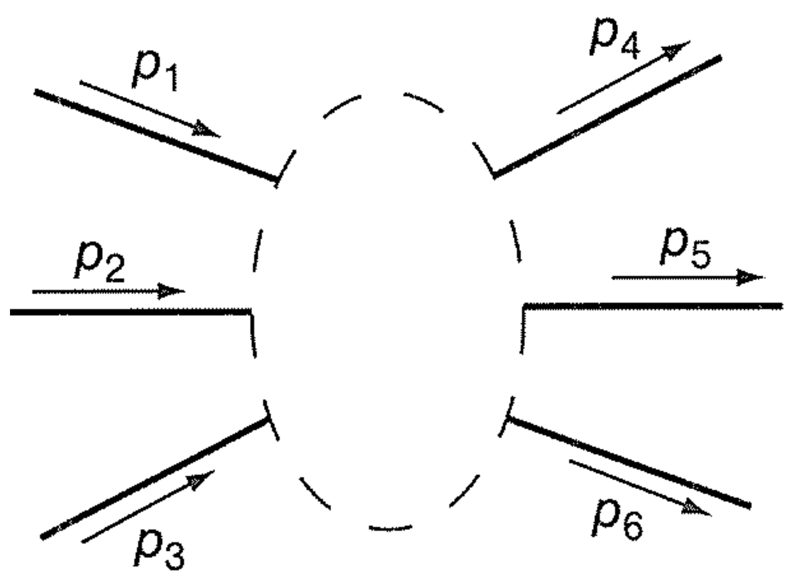

$A + B \to A + B$:




and so on.

Our problem is to find the amplitude $\mathscr{M}$ associated with a given Feynman diagram. The ritual is as follows

1. *Notation*: Label the incoming and outgoing four-momenta $p_1, p_2, \ldots, p_n$ . Label the internal momenta $q_1$, $q_2, \ldots$ Put an arrow beside each line, to keep track of the 'positive' direction (forward in time for external lines, arbitrary for internal lines).
2. *Vertex factors*: For each vertex, write down a factor




$$-ig$$

$g$ is called the *coupling constant*; it specifies the strength of the interaction between $A$, $B$, and $C$. In this toy theory, $g$ has the dimensions of momentum; in the 'real-world' theories, we shall encounter later on, the coupling constant is always dimensionless.

# The Feynman Calculus

## Feynman Rules

3. *Propagators*: For each internal line, write a factor

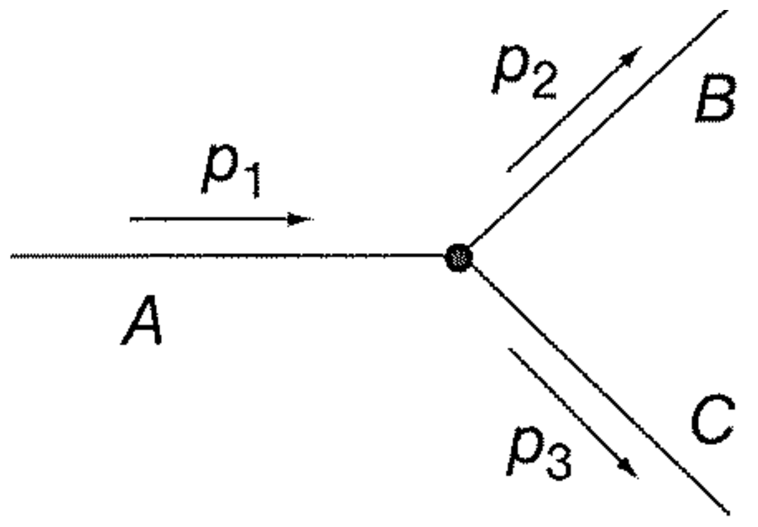
$$\frac{i}{q_j^2 - m_j^2 c^2}$$

   where $q_j$ is the four-momentum of the line and $m_j$ is the mass of the particle the line describes. (Note that $q_j^2 \neq m_j^2 c^2$, because a virtual particle does not lie on its mass shell.)

4. *Conservation of energy and momentum*: For each vertex, write a delta function of the form

$$(2\pi)^4 \delta^4(k_1 + k_2 + k_3)$$

   where the $k$'s are the three four-momenta coming *into* the vertex (if the arrow leads outward, then $k$ is *minus* the four-momentum of that line). This factor imposes conservation of energy and momentum at each vertex, since the delta function is zero unless the sum of the incoming momenta equals the sum of the outgoing momenta.




5. *Integration over internal momenta*: For each internal line, write down a factor*

$$\frac{1}{(2\pi)^4} \, d^4 q_j$$

   and integrate over all internal momenta.

6. *Cancel the delta function*: The result will include a delta function

$$(2\pi)^4 \delta^4(p_1 + p_2 + \cdots - p_n)$$

   reflecting overall conservation of energy and momentum. Erase this factor† and multiply by $i$. The result is $\mathscr{M}$.

### Lifetime of the $A$

The simplest possible diagram, representing the lowest-order contribution to $A \to B + C$, has no internal lines at all . There is one vertex, at which we pick up a factor of $-ig$ (Rule 2) and a delta function

$$(2\pi)^4 \delta^4(p_1 - p_2 - p_3)$$

(Rule 4), which we promptly discard (Rule 6). Multiplying by $i$, we get

$$\mathscr{M} = g$$

This is the *amplitude* (to lowest order); the decay rate is

$$\Gamma = \frac{g^2 |\mathbf{p}|}{8\pi \hbar m_A^2 c} \qquad |\mathbf{p}| = \frac{c}{2m_A}\sqrt{m_A^4 + m_B^4 + m_C^4 - 2m_A^2 m_B^2 - 2m_A^2 m_C^2 - 2m_B^2 m_C^2} \qquad \tau = \frac{1}{\Gamma} = \frac{8\pi \hbar m_A^2 c}{g^2 |\mathbf{p}|}$$

# The Feynman Calculus



**$A + A \to B + B$ Scattering**

The lowest-order contribution to the process $A + A \to B + B$
In this case, there are two vertices (hence two factors of $-ig$), one internal line, with the propagator

$$\frac{i}{q^2 - m_C^2 c^2}$$

two delta functions:

$$(2\pi)^4 \delta^4(p_1 - p_3 - q) \quad and \quad (2\pi)^4 \delta^4(p_2 + q - p_4)$$

and one integration:

$$\frac{1}{(2\pi)^4} d^4 q$$

Rules 1–5, then, yield

$$-i(2\pi)^4 g^2 \int \frac{1}{q^2 - m_C^2 c^2} \delta^4(p_1 - p_3 - q)\, \delta^4(p_2 + q - p_4)\, d^4 q$$

Doing the integral, the second delta function sends $q \to p_4 - p_2$, and we have

$$-ig^2 \frac{1}{(p_4 - p_2)^2 - m_C^2 c^2} (2\pi)^4 \delta^4(p_1 + p_2 - p_3 - p_4)$$

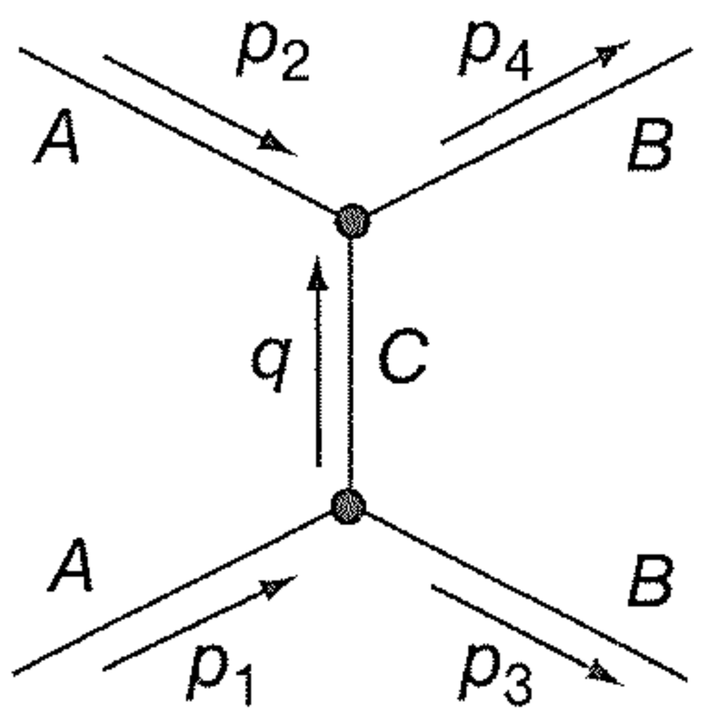
As promised, there is one remaining delta function, reflecting overall conservation of energy and momentum. Erasing it and multiplying by $i$ (Rule 6), we are left with

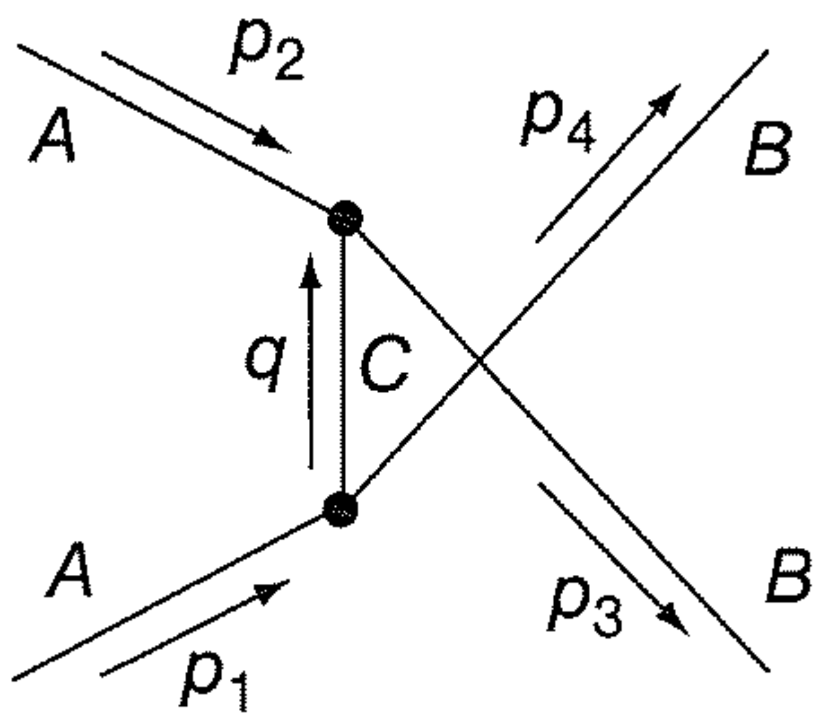
$$\mathscr{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2}$$

But that's not the whole story, for there is another diagram of order $g^2$, obtained by 'twisting' the $B$ lines

$$\mathscr{M} = \frac{g^2}{(p_4 - p_2)^2 - m_C^2 c^2} + \frac{g^2}{(p_3 - p_2)^2 - m_C^2 c^2}$$

Notice, incidentally, that $\mathscr{M}$ is a Lorentz-invariant (scalar) quantity. This is *always* the case; it is built into the Feynman rules.



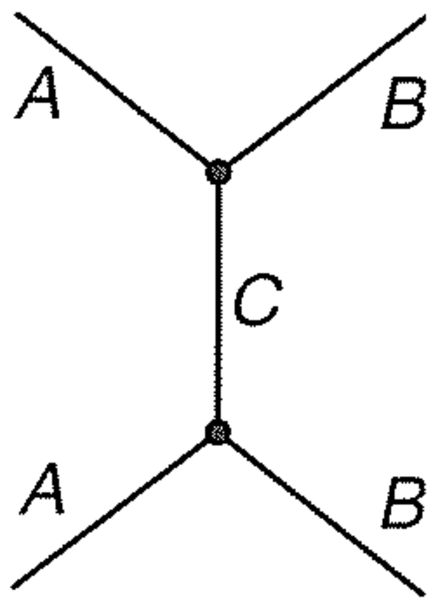
# The Feynman Calculus

## Feynman Rules

Suppose we are interested in the differential cross section $(d\sigma/d\Omega)$ for this process, in the CM system Say, for simplicity, that $m_A = m_B = m$ and $m_C = 0$. Then

$$(p_4 - p_2)^2 - m_C^2 c^2 = p_4^2 + p_2^2 - 2p_2 \cdot p_4 = -2\mathbf{p}^2(1 - \cos\theta)$$

$$(p_3 - p_2)^2 - m_C^2 c^2 = p_3^2 + p_2^2 - 2p_3 \cdot p_2 = -2\mathbf{p}^2(1 + \cos\theta)$$

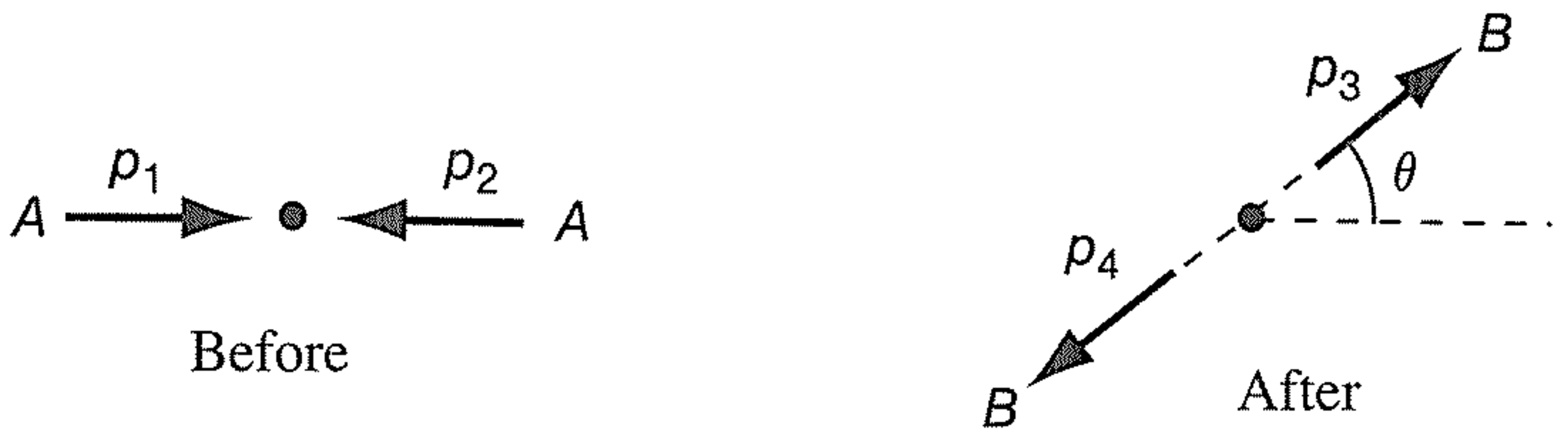



(where $\mathbf{p}$ is the incident momentum of particle 1), and hence

$$\mathcal{M} = -\frac{g^2}{\mathbf{p}^2 \sin^2\theta}$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{2}\left(\frac{\hbar c g^2}{16\pi E\,\mathbf{p}^2 \sin^2\theta}\right)^2$$

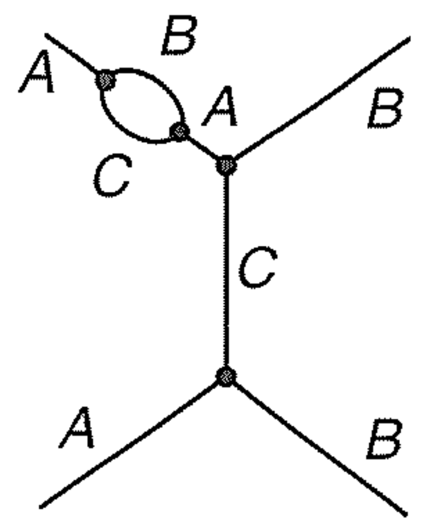
(there are two identical particles in the final state, so $S = 1/2$). As in the case of



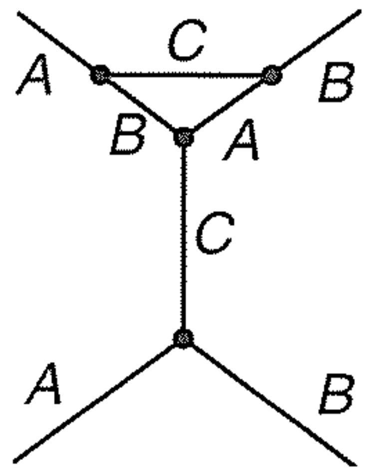

## Higher-order Diagrams

This diagram has two vertices, so $\mathcal{M}$ is proportional to $g^2$. But there are eight diagrams with *four* vertices (and eight more with the external $B$ lines 'twisted'):

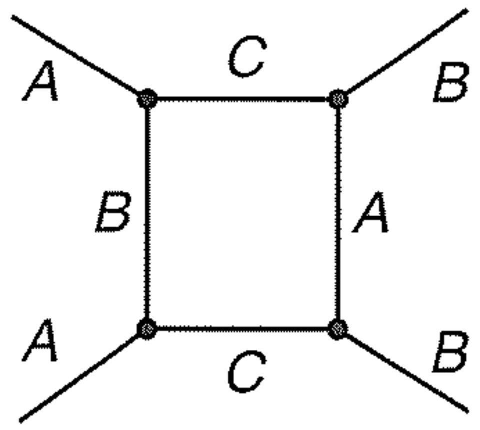
- five 'self-energy' diagrams, in which one of the lines sprouts a loop:




- two 'vertex corrections', in which a vertex becomes a triangle:



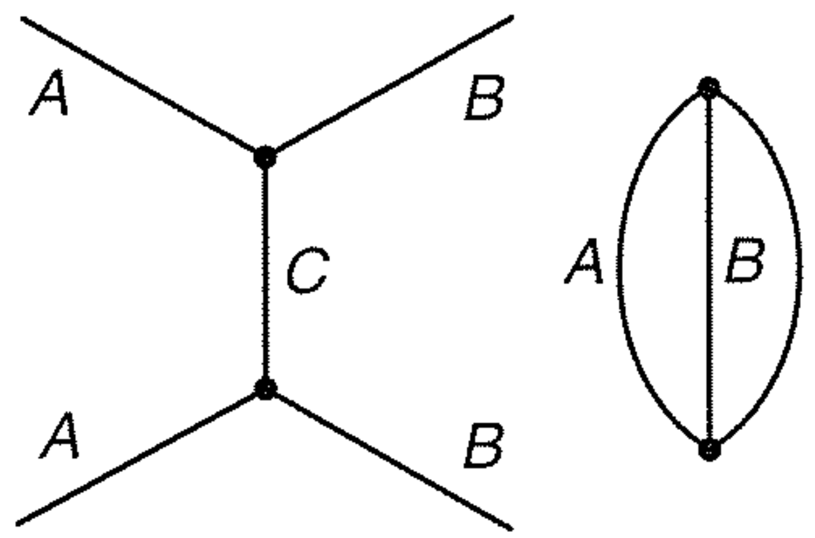

- and one 'box' diagram:



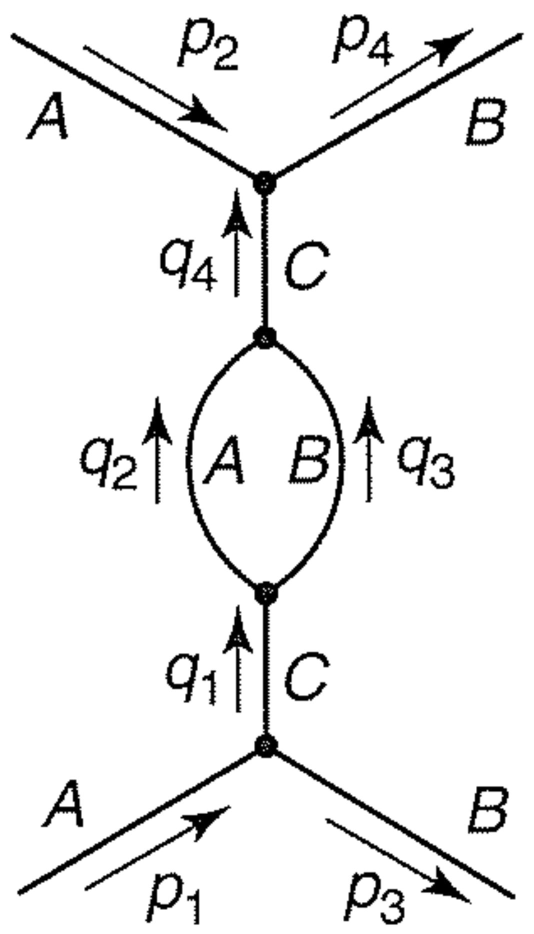
# The Feynman Calculus



(Disconnected diagrams, such as




don't count.)




Applying Feynman rules 1–5, we obtain

$$g^4 \int \frac{\delta^4(p_1 - q_1 - p_3)\,\delta^4(q_1 - q_2 - q_3)\,\delta^4(q_2 + q_3 - q_4)\,\delta^4(q_4 + p_2 - p_4)}{(q_1^2 - m_C^2c^2)(q_2^2 - m_A^2c^2)(q_3^2 - m_B^2c^2)(q_4^2 - m_C^2c^2)}$$
$$\times d^4q_1\, d^4q_2\, d^4q_3\, d^4q_4$$

Integration over $q_1$, using the first delta function, replaces $q_1$ by $(p_1 - p_3)$; integration over $q_4$, using the last delta function, replaces $q_4$ by $(p_4 - p_2)$:

$$\frac{g^4}{[(p_1 - p_3)^2 - m_C^2c^2][(p_4 - p_2)^2 - m_C^2c^2]}$$
$$\times \int \frac{\delta^4(p_1 - p_3 - q_2 - q_3)\,\delta^4(q_2 + q_3 - p_4 + p_2)}{(q_2^2 - m_A^2c^2)(q_3^2 - m_B^2c^2)}\, d^4q_2\, d^4q_3$$

Here, the first delta function sends $q_2 \to p_1 - p_3 - q_3$, and the second delta function becomes

$$\delta^4(p_1 + p_2 - p_3 - p_4)$$

## Higher-order Diagrams

which, by Rule 6, we erase, leaving

$$\mathcal{M} = i\left(\frac{g}{2\pi}\right)^4 \frac{1}{[(p_1 - p_3)^2 - m_C^2c^2]^2} \int \frac{1}{[(p_1 - p_3 - q)^2 - m_A^2c^2](q^2 - m_B^2c^2)}\, d^4q$$

You can try calculating this integral, if you've got the energy, but I'll tell you right now you're going to hit a snag. The four-dimensional volume element could be written as $d^4q = q^3\, dq\, d\Omega'$ (where $d\Omega'$ stands for the angular part), just as in two-dimensional polar coordinates the element of area is $r\, dr\, d\theta$ and in three-dimensional spherical coordinates the volume element is $r^2\, dr \sin\theta\, d\theta\, d\phi$. At large $q$ the integrand is essentially just $1/q^4$, so the $q$ integral has the form

$$\int^{\infty} \frac{1}{q^4} q^3\, dq = \ln q|^{\infty} = \infty$$

The integral is logarithmically divergent at large $q$. This disaster, in one form or another, held up the development of quantum electrodynamics for nearly two decades, until, through the combined efforts of many great physicists – from Dirac, Pauli, Kramers, Weisskopf, and Bethe through Tomonaga, Schwinger, and Feynman – systematic methods were developed for 'sweeping the infinities under the rug'. The first step is to *regularize* the integral, using a suitable cutoff procedure that renders it finite without spoiling other desirable features (such as Lorentz invariance). this can be accomplished by introducing a factor

$$\frac{-M^2c^2}{(q^2 - M^2c^2)}$$

# The Feynman Calculus

At this point, a miraculous thing happens: all the divergent, $M$-dependent terms appear in the final answer in the form of *additions to the masses and the coupling constant*. If we take this seriously, it means that the *physical* masses and couplings are not the $m$'s and $g$'s that appeared in the original Feynman rules, but rather the 'renormalized' ones, containing these extra factors:

$$m_{\text{physical}} = m + \delta m; \quad g_{\text{physical}} = g + \delta g$$

The fact that $\delta m$ and $\delta g$ are infinite (in the limit $M \to \infty$) is disturbing, but not catastrophic, for we never measure them anyway; all we ever see in the laboratory are the *physical* values, and these are (obviously) finite (evidently the unmeasurable 'bare' masses and couplings, $m$ and $g$, contain compensating infinities).* As a practical matter, we take account of the infinities by using the *physical* values of $m$ and $g$ in the Feynman rules, and then systematically ignoring the divergent contributions from higher-order diagrams.

Meanwhile, there remain the *finite* ($M$-*in*dependent) contributions from the loop diagrams. They, too, lead to modifications in $m$ and $g$ (perfectly calculable ones, in this case) – which, however, are functions of the four-momentum of the line in which the loop is inserted ($p_1 - p_3$ in the example). This means that the *effective* masses and coupling constants actually depend on the *energies* of the particles involved; we call them 'running' masses and 'running' coupling constants. The dependence is typically rather slight, at low energies, and can ordinarily be ignored, but it does have observable consequences, in the form of the Lamb shift (in QED) and asymptotic freedom (in QCD).†

# Renormalization